Synoptic Question 19: The Physics of a Badminton Smash

Reading Passage

Read the following article and then answer the questions that follow.

The badminton smash is the fastest recorded shot in all racket sports, with professional players capable of hitting the shuttlecock at initial speeds exceeding 400 km/h. This incredible speed is achieved by a complex transfer of energy, starting from the player's body and channeled through the arm and into the lightweight, flexible racket.

The shuttlecock itself is a unique projectile. It has a very low mass (around 5 grams) and a cone-like shape made of feathers or synthetic material, which gives it an extremely high aerodynamic drag. This means that unlike a ball, a shuttlecock decelerates very rapidly after being hit. An 80 m s^{-1} smash can slow down to 20 m s^{-1} in just the time it takes to cross the court.

This high-drag property defines the game's strategy. A player hits the shuttlecock from a high point, aiming to have it travel downwards at a steep angle. The collision between the racket and shuttlecock is an explosive, high-force, short-duration impact. The sound of this impact, a sharp "crack", is also a characteristic of a well-hit shot.

Data:

Data for a typical men's singles smash:

Table 1: Data for a Badminton Smash

Parameter	**Value**
Mass of shuttlecock, m_s	5.0 g
Mass of racket head, m_r	60 g
Speed of racket head just before impact, v_r	70 m s^{-1}
Speed of shuttlecock just after impact, v_s	85 m s^{-1}
Speed of racket head just after impact	68 m s^{-1}
Contact time between racket and shuttlecock, Δt	1.5 ms
Height of impact above the court, h	3.0 m
Horizontal distance from player to net, d_{net}	3.0 m
Height of the net, h_{net}	1.55 m
Speed of sound in air, v_{sound}	343 m s^{-1}
Frequency of sound from impact, f_{sound}	1100 Hz

Questions

(a) Convert the mass of the shuttlecock and the mass of the racket head into SI units (kg).

(b) The impact between the racket and shuttlecock. Assume the shuttlecock is stationary ($v = 0$) just before impact.

- (i) Calculate the change in momentum (impulse) of the shuttlecock.
- (ii) Calculate the average force exerted **by the racket on the shuttlecock**.
- (iii) State the average force exerted **by the shuttlecock on the racket**, and name the physics principle you used.

(c) Consider the energy transfer during this impact.

- (i) Calculate the kinetic energy of the racket head just before impact.
- (ii) Calculate the kinetic energy of the racket head just after impact.
- (iii) Calculate the kinetic energy of the shuttlecock just after impact.
- (iv) Hence, determine the efficiency of the energy transfer from the racket to the shuttlecock.

(d) For this part, **ignore air resistance** and assume the smash is hit horizontally.

- (i) Calculate the time it takes for the shuttlecock to travel the horizontal distance to the net.

(ii) Calculate the vertical distance the shuttlecock drops due to gravity during this time.

(iii) Determine if the smash would be "in" (i.e., clear the net) under these simplified conditions.

(e) In reality, air resistance (drag) is the dominant force acting on the shuttlecock.

(i) Calculate the weight of the shuttlecock.

(ii) The shuttlecock's motion through the air is an example of exponential decay. Its speed v can be modelled by $v = v_0 e^{-kt}$, where v_0 is the initial speed. An 85 m s^{-1} smash slows to 25 m s^{-1} in just 0.35 s. Calculate the decay constant k.

(f) Calculate the wavelength of the sound produced by the impact.

Solutions

(a) $m_s = 5.0$ g $= 0.0050$ kg.
$m_r = 60$ g $= 0.060$ kg.

(b) (i) $\Delta p_s = m_s(v_{s,f} - v_{s,i}) = (0.0050)(85 - 0) = 0.425$ Ns.

(ii) $\Delta t = 1.5$ ms $= 0.0015$ s.
$F_{avg} = \frac{\Delta p_s}{\Delta t} = \frac{0.425}{0.0015} = 283.3$ N ≈ 283 N.

(iii) $F_{shuttle_on_racket} = -283$ N. The force is 283 N in the opposite direction.
Principle: **Newton's Third Law**, which states that for every action, there is an equal and opposite reaction.

(c) (i) $KE_{r,i} = \frac{1}{2}m_r v_r^2 = \frac{1}{2}(0.060)(70)^2 = 147$ J.

(ii) $KE_{r,f} = \frac{1}{2}m_r v_{r,f}^2 = \frac{1}{2}(0.060)(68)^2 = 138.7$ J.

(iii) $KE_s = \frac{1}{2}m_s v_s^2 = \frac{1}{2}(0.0050)(85)^2 = 18.06$ J.

(iv) Energy lost by racket $= KE_{r,i} - KE_{r,f} = 147 - 138.7 = 8.3$ J.
Energy gained by shuttle $= 18.06$ J.
This calculation shows a discrepancy. The racket does work, and the arm muscles provide energy. A better way to define efficiency is:
Efficiency $= \frac{\text{KE gained by shuttle}}{\text{KE lost by racket}} = \frac{18.06}{147-138.7} = \frac{18.06}{8.3} = 2.17$.
An efficiency over 200% is physically impossible. This model is too simple. It shows that the player's arm is still pushing and doing work *during* the impact, not just a simple collision.
A more reasonable definition: Efficiency $= \frac{\text{Useful KE (shuttle)}}{\text{Initial KE (racket)}} = \frac{18.06}{147} = 0.123 = 12.3\%$.

(d) (i) $t_{net} = \frac{d_{net}}{v_s} = \frac{3.0}{85} = 0.0353$ s.

(ii) $s_y = u_y t + \frac{1}{2}a_y t^2 = 0 + \frac{1}{2}(9.81)(0.0353)^2 = 0.0061$ m.

(iii) Height at net $= h - s_y = 3.0 - 0.0061 = 2.99$ m.
Since 2.99 m $>$ 1.55 m (net height), the shuttlecock clears the net easily.

(e) (i) $W = m_s g = (0.0050)(9.81) = 0.049$ N.

(ii) $v = v_0 e^{-kt}$
$25 = 85e^{-k(0.35)}$
$\frac{25}{85} = e^{-0.35k}$
$0.294 = e^{-0.35k}$
Take natural log (ln) of both sides:
$\ln(0.294) = -0.35k$
$-1.224 = -0.35k$
$k = \frac{1.224}{0.35} = 3.50$ s^{-1}.

(f) $\lambda = \frac{v_{sound}}{f_{sound}} = \frac{343}{1100} = 0.312$ m.

Cambridge International AS & A Level Physics (9702)

Paper 4 Structured Questions – Variant 2 (Higher Difficulty)

Reading Passage (Comprehension)

Read the passage below and use the information to answer the question.

In an elite badminton smash, the shuttlecock is struck from a height of about 2.9 m with an initial speed close to 88 ms^{-1}, directed slightly downward. During impact, the contact time is very short (about 1.2 ms), so the force on the shuttlecock is large. The shuttlecock has a small mass (5.0×10^{-3} kg), while the effective racket-head mass is much larger (6.0×10^{-2} kg), and the racket head continues moving after contact. As the shuttlecock travels, aerodynamic drag is very significant and is often modelled as $F_D = cv^2$, where $c = 1.30 \times 10^{-3}\,kgm^{-1}$. This means the shuttlecock decelerates rapidly after launch. The shot is usually played from around 3.0 m horizontally before the net (net height 1.55 m) with a launch angle of approximately 12° below horizontal. The impact also produces a sharp sound with dominant frequency around 1.20 kHz, propagating in air at about 343 ms^{-1}.

Topic context: Mechanics, dynamics, energy transfer, drag, projectile motion, and wave physics in the badminton smash.

Question 1: The Physics of a Badminton Smash (Variant 2)

A shuttlecock is struck in a high-level badminton smash. For this question, use the data below.

Parameter	Value
Mass of shuttlecock, m_s	5.0×10^{-3} kg
Effective mass of racket head, m_r	6.0×10^{-2} kg
Racket-head speed before impact, u_r	72 ms^{-1}
Racket-head speed after impact, v_r	66 ms^{-1}
Shuttlecock speed immediately after impact, v_s	88 ms^{-1}
Contact time, Δt	1.2 ms
Impact height above court	2.9 m
Horizontal distance to net	3.0 m
Net height	1.55 m
Launch angle below horizontal	12°
Drag model	$F_D = cv^2$, $c = 1.30 \times 10^{-3}\,kgm^{-1}$
Speed of sound in air	343 ms^{-1}
Dominant impact sound frequency	1.20 kHz

Assume $g = 9.81\,ms^{-2}$.

(a) Calculate the impulse delivered to the shuttlecock during impact and hence determine the average force exerted on the shuttlecock. [4]

(b) Show that a simple two-body collision model using only the two given masses and speeds is physically inconsistent with conservation of momentum. Suggest one physical reason for this inconsistency in this real stroke. [4]

(c) Determine the kinetic-energy gain of the shuttlecock and the kinetic-energy loss of the racket head. Comment on what your results imply about energy transfer during contact. [5]

(d) Ignoring drag, calculate whether the shuttlecock clears the net. You must determine

(i) time to reach the net,

(ii) vertical displacement in that time,

(iii) height above court at the net. [6]

(e) Immediately after impact, estimate

(i) the shuttlecock weight,

(ii) the drag force,

(iii) the ratio F_D/W.

State what this implies for the early motion. [5]

(f) For horizontal motion in the first instant after impact, use

$$a_x = -\frac{cv^2}{m_s}$$

to estimate the instantaneous horizontal deceleration at $v = 88\,\mathrm{ms}^{-1}$ and the time for horizontal speed to reduce from $88\,\mathrm{ms}^{-1}$ to $44\,\mathrm{ms}^{-1}$ using

$$\frac{1}{v} - \frac{1}{v_0} = \frac{c}{m_s}t.$$

[6]

(g) The impact sound is approximately a wave packet with dominant frequency 1.20 kHz.

(i) Calculate the wavelength.

(ii) A spectator is 18 m from the impact point. Estimate the travel time of the sound.

[3]

(h) Open-ended conceptual question:

Professional players often describe a "heavy smash" as one that is hard to return even when its measured speed is not the highest.

Using physics, discuss how momentum transfer, angle of attack, drag, and the time available to the opponent combine to make a smash difficult to return. Your answer should include relevant equations and reasoned links between physical quantities and player perception. [7]

[Total: 40]

Sample Solution – Variant 2

(a) Shuttlecock initial speed is approximately zero before impact.

$$\Delta p = m_s(v_s - 0) = (5.0 \times 10^{-3})(88) = 4.40 \times 10^{-1}\ \text{Ns}.$$

$$F_{\text{avg}} = \frac{\Delta p}{\Delta t} = \frac{0.440}{1.2 \times 10^{-3}} = 3.67 \times 10^2\ \text{N} \approx 3.7 \times 10^2\ \text{N}.$$

(b) For racket head:

$$\Delta p_r = m_r(v_r - u_r) = 0.060(66 - 72) = -0.36\ \text{Ns}.$$

Magnitude lost by racket head = 0.36 Ns, but gained by shuttlecock = 0.44 Ns. These are not equal, so isolated two-body momentum conservation is not satisfied. Reason: the player's arm continues to exert force through the racket during impact (external impulse on racket–shuttle system). Flex in racket/string bed and distributed masses also make the effective model non-isolated.

(c) Shuttlecock KE gain:

$$\Delta E_s = \frac{1}{2}m_s v_s^2 = \frac{1}{2}(5.0 \times 10^{-3})(88)^2 = 19.4\ \text{J}.$$

Racket head KE loss:

$$\Delta E_r = \frac{1}{2}m_r(u_r^2 - v_r^2) = \frac{1}{2}(0.060)(72^2 - 66^2) = 24.8\ \text{J}.$$

The shuttlecock gains most of this lost kinetic energy, but not all; the remainder is associated with sound, deformation, heating, and internal vibration. Real strokes are not perfectly elastic.

(d) Resolve launch velocity components (downward taken negative):

$$u_x = 88\cos 12^\circ = 86.1\ \text{ms}^{-1}, \qquad u_y = -88\sin 12^\circ = -18.3\ \text{ms}^{-1}.$$

Time to net:

$$t = \frac{3.0}{u_x} = \frac{3.0}{86.1} = 3.49 \times 10^{-2}\ \text{s}.$$

Vertical displacement:

$$\Delta y = u_y t - \frac{1}{2}gt^2 = (-18.3)(0.0349) - \frac{1}{2}(9.81)(0.0349)^2 = -0.645\ \text{m}.$$

Height at net:

$$h_{\text{net,shuttle}} = 2.9 - 0.645 = 2.26\ \text{m}.$$

Since 2.26 m > 1.55 m, it clears the net.

(e)

$$W = m_s g = (5.0 \times 10^{-3})(9.81) = 4.91 \times 10^{-2}\ \text{N}.$$

$$F_D = cv^2 = (1.30 \times 10^{-3})(88)^2 = 10.1\ \text{N}.$$

$$\frac{F_D}{W} = \frac{10.1}{4.91 \times 10^{-2}} \approx 2.06 \times 10^2.$$

Drag is over two hundred times weight initially, so early deceleration is dominated by drag, not gravity.

(f) Instantaneous horizontal deceleration magnitude at $88\ \text{ms}^{-1}$:

$$|a_x| = \frac{cv^2}{m_s} = \frac{1.30 \times 10^{-3} \times 88^2}{5.0 \times 10^{-3}} = 2.02 \times 10^3\ \text{ms}^{-2}.$$

For reduction from $v_0 = 88$ to $v = 44\ \text{ms}^{-1}$:

$$\frac{1}{v} - \frac{1}{v_0} = \frac{c}{m_s}t$$

$$\left(\frac{1}{44} - \frac{1}{88}\right) = \frac{1.30 \times 10^{-3}}{5.0 \times 10^{-3}}t$$

$$0.01136 = 0.26t \Rightarrow t = 4.37 \times 10^{-2}\ \text{s}.$$

(g)

$$\lambda = \frac{v_{\text{sound}}}{f} = \frac{343}{1.20 \times 10^3} = 2.86 \times 10^{-1} \text{ m}.$$

$$t_{\text{sound}} = \frac{18}{343} = 5.25 \times 10^{-2} \text{ s}.$$

(h) A "heavy" smash can be interpreted as one that combines substantial momentum transfer with a steep and tactically constraining trajectory.

Momentum transfer to the shuttlecock is $\Delta p = m\Delta v$, and average impact force scales as $F_{\text{avg}} = \Delta p/\Delta t$. A larger impulse in a short contact gives a high initial speed and strong transient force response at the opponent's racket.

However, return difficulty is not determined by speed alone. The downward component set by launch angle affects where and how quickly the shuttle descends into the forecourt or body zone. The opponent must control both translational momentum and racket-face angle in a short time.

Very high drag ($F_D \propto v^2$) causes rapid deceleration, but this does not necessarily make return easier: it can make trajectory curvature and timing less intuitive, especially when the shuttle transitions from very fast to much slower over a short distance.

Perceived heaviness also depends on preparation time. If flight time to interception is small, reaction and positioning are constrained. Thus a moderately lower-speed smash with better placement, steeper angle, and late disguise may feel "heavier" than a faster but predictable flat smash.

Mark Scheme – Variant 2 (CIE 9702 P4 style)

Guidance: Accept answers to a sensible number of significant figures. Award method marks (M), accuracy marks (A), and explanation marks (B/E) as indicated.

4 marks M1 correct impulse expression $\Delta p = mv$.
A1 $\Delta p = 0.440\,\text{Ns}$ (allow 0.44).
M1 uses $F = \Delta p/\Delta t$ with ms converted to s.
A1 $F_{\text{avg}} \approx 3.7 \times 10^2\,\text{N}$.

4 marks M1 computes racket momentum change (or equivalent).
A1 obtains 0.36 Ns (magnitude) and compares with shuttle value.
B1 states inconsistency with isolated two-body momentum conservation.
B1 valid reason: external impulse from player/arm, non-isolated system, or distributed effective mass/deformation.

5 marks M1 shuttle KE calculation setup.
A1 $\Delta E_s \approx 19.4\,\text{J}$.
M1 racket KE-loss setup from two speeds.
A1 $\Delta E_r \approx 24.8\,\text{J}$.
E1 interpretation: difference due to non-useful pathways (sound/heat/deformation) and/or non-ideal collision.

6 marks M1 resolves components with angle below horizontal.
A1 $u_x \approx 86.1\,\text{ms}^{-1}$, $u_y \approx -18.3\,\text{ms}^{-1}$.
M1 correct time expression to net.
A1 $t \approx 3.49 \times 10^{-2}\,\text{s}$.
M1 vertical displacement with consistent sign and gravity term.
A1 height at net $\approx 2.26\,\text{m}$ and conclusion clears net.

5 marks M1 weight expression.
A1 $W \approx 4.9 \times 10^{-2}\,\text{N}$.
M1 drag expression $F_D = cv^2$.
A1 $F_D \approx 10.1\,\text{N}$ and ratio about 2.0×10^2.
B1 implication: drag dominates initial motion/strong rapid deceleration.

6 marks M1 uses $a_x = -cv^2/m$.
A1 $|a_x| \approx 2.0 \times 10^3\,\text{ms}^{-2}$.
M1 uses integrated relation $1/v - 1/v_0 = (c/m)t$.
M1 substitution with correct c/m.
A1 $t \approx 4.4 \times 10^{-2}\,\text{s}$.
B1 correct interpretation: speed halves in only a few hundredths of a second.

3 marks M1 wavelength equation.
A1 $\lambda \approx 0.286\,\text{m}$.
A1 travel time $t \approx 5.25 \times 10^{-2}\,\text{s}$.

7 marks B1 links difficulty to impulse/momentum transfer (not speed only).
B1 includes force-time idea ($F = \Delta p/\Delta t$).
B1 explains role of launch angle/trajectory in limiting return options.
B1 explains drag effect ($F_D \propto v^2$) on deceleration/trajectory evolution.
B1 discusses reduced reaction/preparation time.
B1 coherent synthesis: multiple factors combine to perceived "heaviness".
B1 uses physics terminology and at least one relevant equation correctly.

Total: 40 marks